BEACON

A Low-Cost SOS Wearable with Satellite Fallback for Off-Grid Emergency Response

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beaconsos.com

1 ABSTRACT

Emergencies such as assaults, falls, crashes and sudden medical events demand help within seconds, yet nearly every personal safety tool assumes a working, unlocked phone in your hand. Phones die, get taken, or cannot be used under stress or after a fall. Beacon is a one-button wearable with its own cellular link, GPS and a planned satellite fallback: one press sends the user's location and medical information to their contacts and campus police, with no phone and no monthly fee. This project tests whether such a device can be built from commercial cellular-IoT hardware, deliver an SOS end to end, and stay at consumer cost.

~75% [5]	**1 in 4** [2]	**23.3** [1]
of people freeze or act ineffectively in a sudden emergency	U.S. adults 65 and older fall each year	violent victimizations per 1,000 people age 12+ (2024)

- **Hypothesis 1 (H1), standalone:** one press reaches contacts over the device's own LTE-M link, with no phone.
- **Hypothesis 2 (H2), satellite:** 3GPP Release-17 NB-NTN on the nRF9151 carries the same short alert where there is no cell coverage.
- **Hypothesis 3 (H3), cost:** hardware plus five years of connectivity fits a $50–$100 one-time price.

2 DEVICE DESIGN



Figure 1. Two concept enclosure designs at the same scale (40–44 mm, one button, LED confirmation).

Tier	Price	Adds
Lite	$50	Own LTE-M + GPS; Bluetooth for setup only
Scout	$75	+ fall detection, live audio to responders
Global	$100	+ NB-NTN satellite fallback



Watch
BEST FOR: SPORTS, EVERYDAY



Keychain
BEST FOR: KEYS, BAGS



Pendant
BEST FOR: DISCREET WEAR



Belt clip
BEST FOR: OUTDOOR, TRAVEL



PopSocket
BEST FOR: COMMUTERS



Phone case
BEST FOR: TRAVELERS

Figure 2. One core module, six mounts (concept renders); all work without a phone.

Cost compared with existing devices

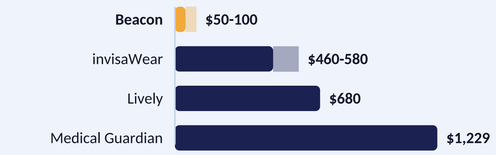


Figure 3. Approximate 2-year cost in US$, device plus required subscription [4], [6], [7], [10].

3 METHODS: SYSTEM DESIGN

I designed the full stack myself: C firmware on Zephyr RTOS (nRF Connect SDK), a custom PCB in Flux, and an AWS backend (Lambda and DynamoDB) that stores user data with no third-party sharing. Prototyping is on the Nordic nRF9160 DK with an iBASIS eSIM; production moves to the nRF9151, which adds NB-NTN satellite support.

Activation: fine motor skills fail under stress, so Beacon fires only on a 3-second hold or rapid double-tap, and a halo LED silently confirms the alert reached the network.

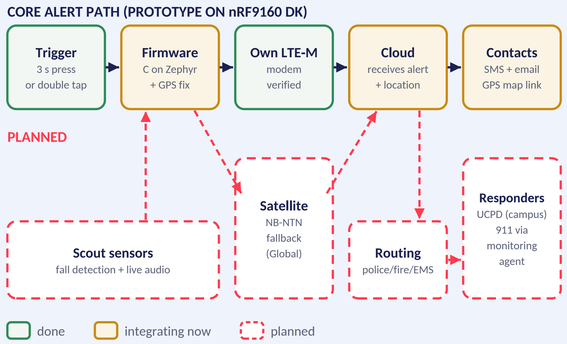


Figure 4. Beacon alert flow and build status.

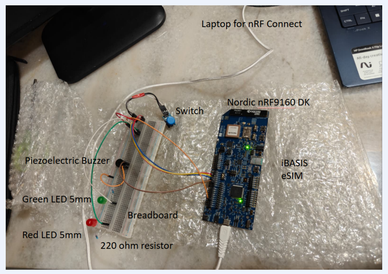


Figure 5. Dev-kit prototype: nRF9160 DK, switch, buzzer, status LEDs and IoT eSIM.

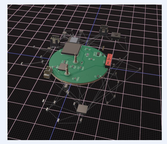


Figure 6. 42 mm circular PCB, 3D board in Flux.

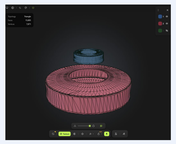


Figure 7. Enclosure 3D model in progress.



Figure 8. Concept cutaway (illustrative).

4 RESULTS

Milestone	Status
Dev-kit hardware assembled (switch, buzzer, LEDs, IoT SIM)	Done
Modem communication verified with nRF Connect (nRF9160)	Done
End-to-end alert: button → LTE-M → cloud → SMS and email	In progress (first test milestone)
Custom 42 mm PCB designed and built	Built; firmware not yet running
Website and dashboard with sample setup (no real user data)	Live
Funding: UChicago Tech Showcase 2026 ($1,250 + $3,000)	**$4,250 non-dilutive**

5 SATELLITE FALLBACK: THE SPACE LAYER

Beacon uses a cell tower first. With no coverage, the Global tier sends the same short SOS **directly to a satellite** over 3GPP Release-17 NB-NTN, from the same Nordic module.

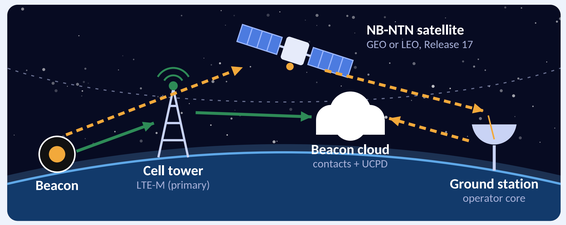


Figure 9. Terrestrial-first, satellite-fallback architecture (planned for the Global tier).

Fallback logic I am implementing

1. **Press; scan LTE-M, none found** — a dwell timer prevents rapid switching
2. **Get a GNSS fix** — NTN attach needs position and timing
3. **Attach to NB-NTN, send short SOS** — a few bytes: ID, GPS, alert type
4. **LED confirms; keep retrying LTE-M** — satellite sends use the most power

Figure 10. Satellite fallback logic.



Figure 11. Illustrative global satellite coverage.

Why the nRF9151

- Release-17 NB-NTN firmware adds satellite to a module that already does LTE-M and GNSS [8].
- Certified on Skylo's commercial NTN network: 36 countries, 60M+ sq km [9].
- **Why not LoRa:** dispatchers do not monitor unlicensed 868/915 MHz bands, and LoRa needs a gateway.

6 USER RESEARCH

More than a dozen interviews and surveys with UChicago students, parents and faculty on how they stay safe, what they carry, and what they would pay.

- **Phones fail at the worst moment:** students' location sharing goes dark when the phone dies or is out of reach.
- **Location is not safety:** a parent called location sharing "a false sense of security." Her biggest worry, a fall at home, shaped Scout's fall detection.
- **It has to go where keys go:** every student in our carry survey brings dorm keys daily, so Beacon mounts on a keychain.
- **Parents buy, but today's tools go unused:** parents bought students pepper spray or alarms that none had ever used. Willingness to pay ranged from a few dollars to $100.

7 ABOUT THE RESEARCHER

UChicago × Fermilab	Undergraduate RA, AI-Enabled Accelerator Operations: agentic AI on FAST/IOTA and PIP-II digital twins
UChicago Space Program	Instrumentation; PULSE-A NASA CubeSat avionics PCBs and laser driver; DOLL-E simulator; rocketry payload; cloud chamber
SkinIO	Built 13 AI agents; worked on Connection to Care, an algorithm routing patients to in-network clinics
Stone Edge Observatory	Research Assistant: bias, dark and flat-field calibration pipelines

8 PROGRESS AND FUTURE WORK

Planned measurements

Measurement	Method	Expected
End-to-end latency (LTE-M)	Repeated presses across campus; log timestamps	Seconds; median and worst case reported
Delivery success rate	Indoor/outdoor trials, LTE-M and NTN	Near 100% on LTE-M; lower on NTN under obstructed sky
Time to satellite lock	GNSS fix, then NTN attach, open sky	Minutes, not seconds [3]
Transmit energy	Power profiler per LTE-M and NTN send	NTN send is the costliest event
Standby current	Sleep current on production board	Supports 2+ years of standby (Lite target)

Expected values are design targets, not yet measured.

Where each hypothesis stands

- **H1, partly supported:** the modem is verified on the dev kit, but the full button-to-contact alert is not yet shown; LTE-M delivery is expected within seconds.
- **H2, untested:** needs the nRF9151. If satellite lock is too slow, the device can keep a recent GNSS fix and start NTN while still scanning for cell coverage.
- **H3, untested:** base cellular hardware is estimated at $24–43 per unit at 10,000 units [10]; connectivity needs written IoT SIM quotes (target ~$2 per device for 5 years).

NOW
Dev-kit prototype assembled; modem verified; PCB built; website live

+2 MONTHS
End-to-end alert; nRF9151 version with satellite and fall detection

+5 MONTHS
UChicago pilot with female students; alerts to UCPD; campus dashboard

+10 MONTHS
Mass production: FCC and carrier certification; bulk university orders

Figure 12. Development roadmap.

Limitations: results so far are build milestones, not measurements, and the user research is a small, UChicago-only sample. The next round expands to 30–40 interviews weighted toward women living near campus.

Significance: if the measurements hold, satellite-backed SOS no longer needs dedicated satellite hardware and can ship in a $100 consumer device.

Welcome to the future of safety.

One button. One message. No phone, no subscription. Help in seconds.

9 REFERENCES

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